

Factor Analysis

KMO and Bartlett's Test

Kaiser-Meyer-Olkin Measure of Sampling Adequacy.		.927
Bartlett's Test of Sphericity	Approx. Chi-Square	10318.490
	df	300
	Sig.	.000

Communalities

	Initial	Extraction
ID1	.548	.638
ID2	.502	.565
ID3	.442	.488
ID4	.449	.513
ID5	.574	.667
EE1	.630	.698
EE2	.616	.681
EE3	.593	.652
EE4	.589	.647
EE5	.601	.662
BG1	.559	.623
BG2	.578	.638
BG3	.574	.620
BG4	.568	.628
BG5	.623	.700
EG1	.497	.564
EG2	.517	.596
EG3	.484	.558
EG4	.483	.545
EG5	.493	.576
CD1	.602	.698
CD2	.545	.604
CD3	.483	.537
CD4	.522	.585
CD5	.532	.596

Extraction Method: Principal Axis
Factoring.

Total Variance Explained

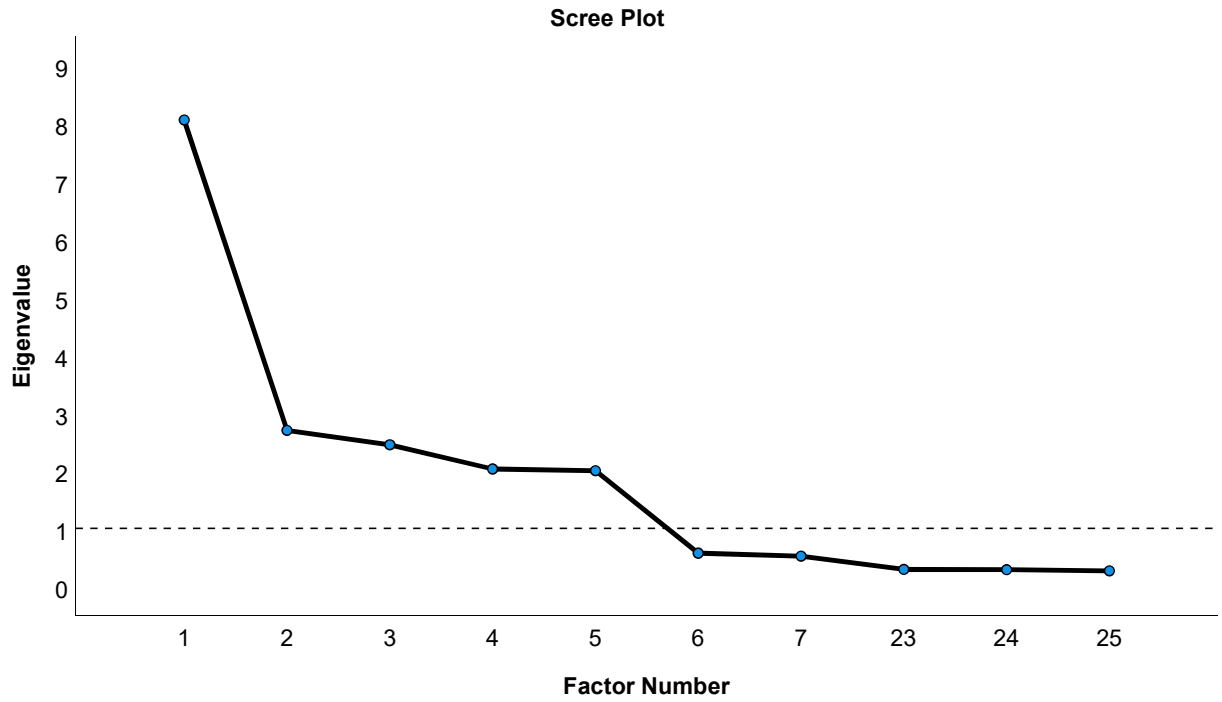
Factor	Initial Eigenvalues			Extraction Sums of Squared Loadings	
	Total	% of Variance	Cumulative %	Total	% of Variance
1	8.057	32.227	32.227	7.673	30.693
2	2.692	10.768	42.995	2.338	9.352
3	2.444	9.777	52.772	2.039	8.156
4	2.025	8.099	60.871	1.652	6.608
5	1.995	7.982	68.853	1.578	6.314
6	.570	2.281	71.133		
7	.518	2.072	73.206		
8	.487	1.947	75.153		
9	.482	1.927	77.080		
10	.451	1.803	78.883		
11	.442	1.770	80.653		
12	.436	1.745	82.398		
13	.421	1.683	84.081		
14	.402	1.607	85.688		
15	.390	1.560	87.248		
16	.369	1.478	88.726		
17	.358	1.433	90.159		
18	.340	1.361	91.520		
19	.332	1.329	92.848		
20	.327	1.306	94.155		
21	.321	1.285	95.440		
22	.304	1.217	96.656		
23	.289	1.154	97.810		
24	.284	1.137	98.948		
25	.263	1.052	100.000		

Total Variance Explained

Factor	Extraction Sums of Squared ...	Rotation Sums of Squared Loadings ^a
	Cumulative %	Total
1	30.693	4.708
2	40.045	5.289
3	48.201	4.876
4	54.809	4.265
5	61.122	4.846
6		
7		
8		
9		
10		
11		
12		
13		
14		
15		
16		
17		
18		
19		
20		
21		
22		
23		
24		
25		

Extraction Method: Principal Axis Factoring.

a. When factors are correlated, sums of squared loadings cannot be added to obtain a total variance.



Factor Matrix^a

	Factor				
	1	2	3	4	5
ID1			.634		
ID2	.511		.530		
ID3			.527		
ID4			.561		
ID5	.517		.619		
EE1	.570	.557			
EE2	.567	.561			
EE3	.534	.567			
EE4	.537	.559			
EE5	.536	.565			
BG1	.595				
BG2	.625				
BG3	.634				
BG4	.616				
BG5	.646				
EG1	.550				

Factor Matrix^a

	Factor				
	1	2	3	4	5
EG2	.567				
EG3	.543				
EG4	.561				
EG5	.523				.508
CD1	.593				
CD2	.590				
CD3	.545				
CD4	.540				
CD5	.532				

Extraction Method: Principal Axis Factoring.

a. 5 factors extracted. 5 iterations required.

Pattern Matrix^a

	Factor				
	1	2	3	4	5
ID1				.809	
ID2				.721	
ID3				.693	
ID4				.728	
ID5				.814	
EE1	.829				
EE2	.814				
EE3	.809				
EE4	.804				
EE5	.816				
BG1		.805			
BG2		.791			
BG3		.756			
BG4		.788			
BG5		.840			
EG1					.748
EG2					.766
EG3					.740

Pattern Matrix^a

	Factor				
	1	2	3	4	5
EG4					.705
EG5					.784
CD1			.836		
CD2			.746		
CD3			.720		
CD4			.779		
CD5			.777		

Extraction Method: Principal Axis Factoring.

Rotation Method: Promax with Kaiser Normalization.^a

a. Rotation converged in 5 iterations.

Structure Matrix

	Factor				
	1	2	3	4	5
ID1				.796	
ID2				.747	
ID3				.698	
ID4				.715	
ID5				.817	
EE1	.835				
EE2	.825				
EE3	.805				
EE4	.804				
EE5	.812				
BG1		.789			
BG2		.798			
BG3		.786			
BG4		.791			
BG5		.837			
EG1					.750
EG2					.772
EG3					.745
EG4					.733

Structure Matrix

	Factor				
	1	2	3	4	5
EG5					.757
CD1			.835		
CD2			.775		
CD3			.728		
CD4			.764		
CD5			.768		

Extraction Method: Principal Axis Factoring.

Rotation Method: Promax with Kaiser Normalization.

Factor Correlation Matrix

Factor	1	2	3	4	5
1	1.000	.370	.310	.318	.340
2	.370	1.000	.457	.357	.460
3	.310	.457	1.000	.316	.447
4	.318	.357	.316	1.000	.337
5	.340	.460	.447	.337	1.000

Extraction Method: Principal Axis Factoring.

Rotation Method: Promax with Kaiser Normalization.

Multilayer Perceptron

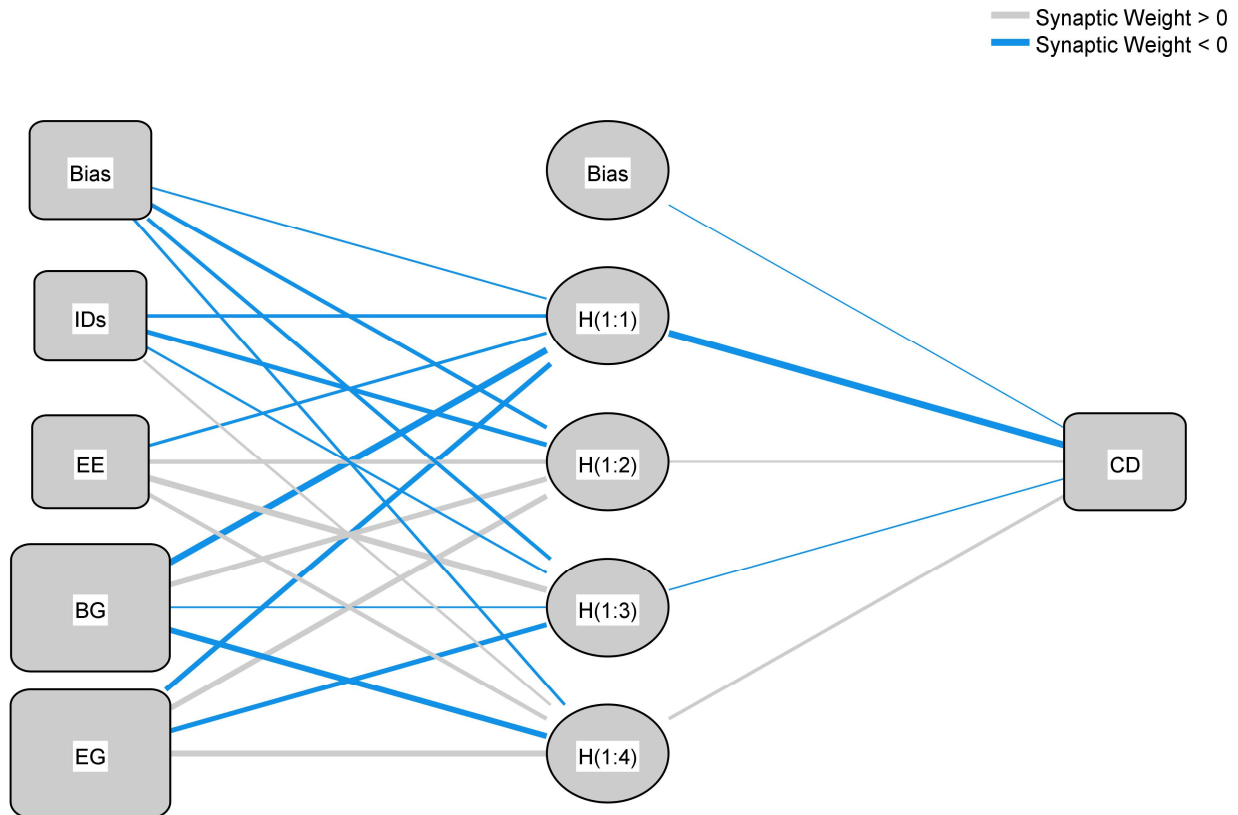
Case Processing Summary

		N	Percent
Sample	Training	513	68.4%
	Testing	237	31.6%
Valid		750	100.0%
Excluded		0	
Total		750	

Network Information

Input Layer	Covariates	1	IDs
		2	EE
		3	BG
		4	EG
	Number of Units ^a		4
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1 ^a		4
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	CD
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit



Hidden layer activation function: Hyperbolic tangent

Output layer activation function: Identity

Model Summary

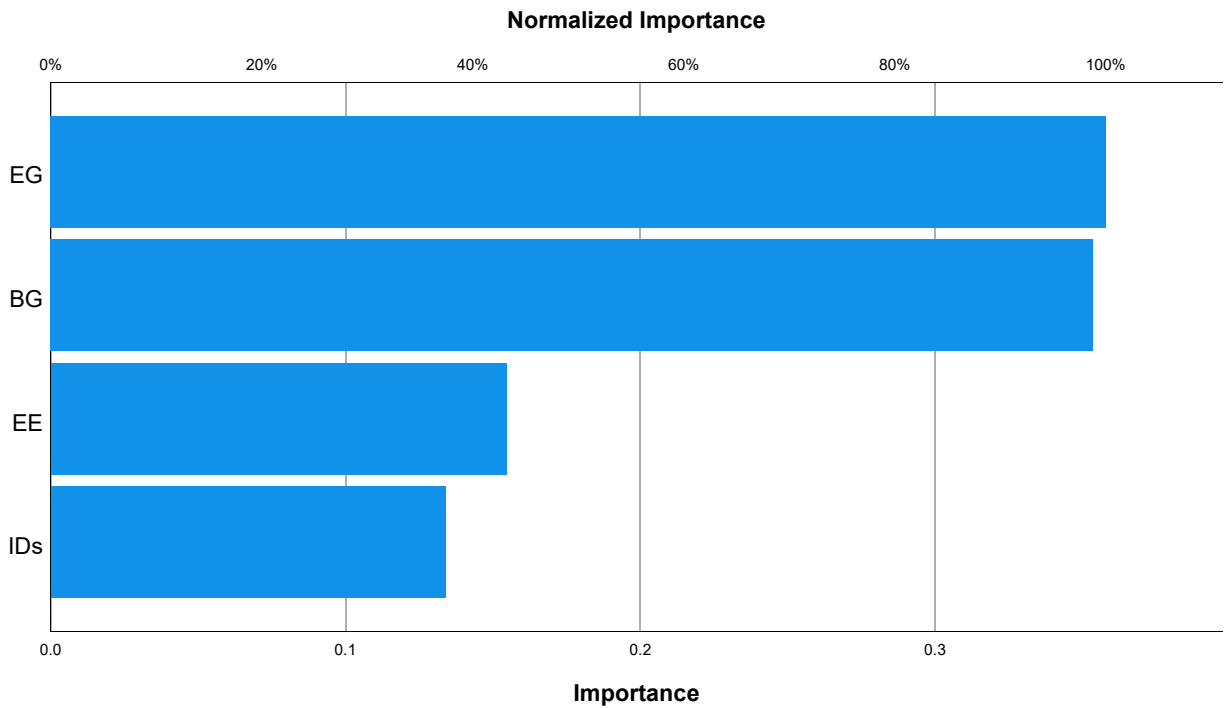
Training	Sum of Squares Error	192.285
	Relative Error	.751
	Stopping Rule Used	1 consecutive step (s) with no decrease in error ^a
	Training Time	0:00:00.05
Testing	Sum of Squares Error	86.607
	Relative Error	.761

Dependent Variable: CD

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
IDs	.134	37.4%
EE	.155	43.2%
BG	.353	98.8%
EG	.358	100.0%



Multilayer Perceptron

Case Processing Summary

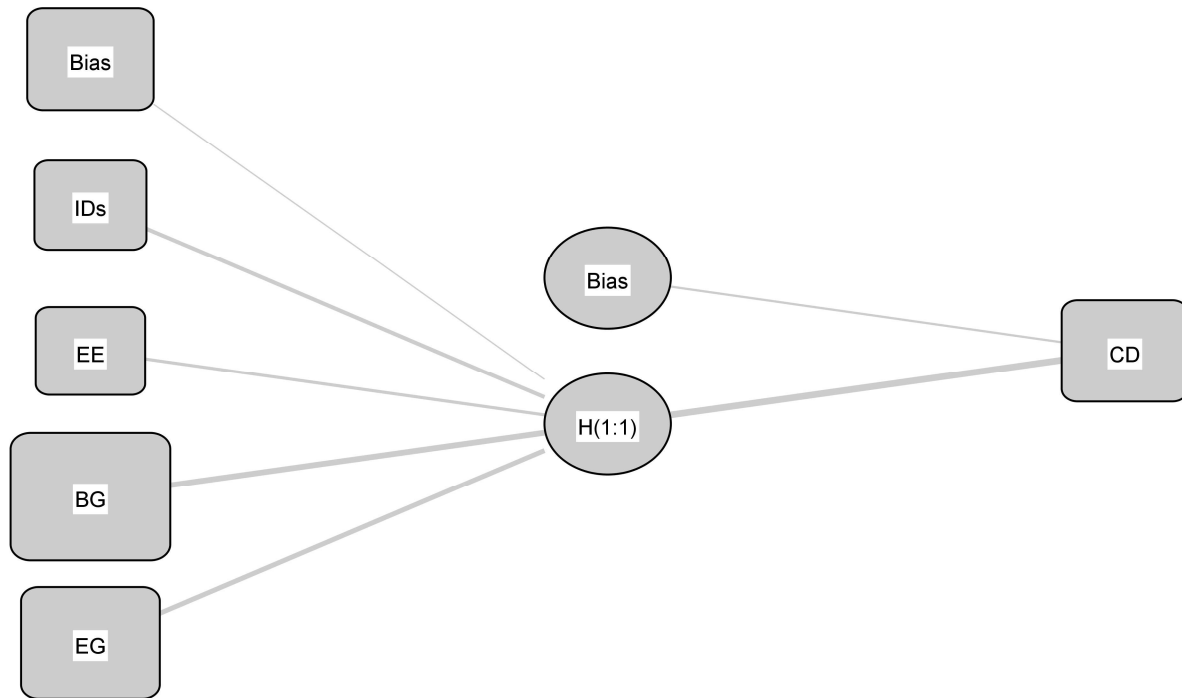
		N	Percent
Sample	Training	526	70.1%
	Testing	224	29.9%
Valid		750	100.0%
Excluded		0	
Total		750	

Network Information

Input Layer	Covariates	1	IDs
		2	EE
		3	BG
		4	EG
	Number of Units ^a		4
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1 ^a		1
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	CD
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit

— Synaptic Weight > 0



Hidden layer activation function: Hyperbolic tangent

Output layer activation function: Identity

Model Summary

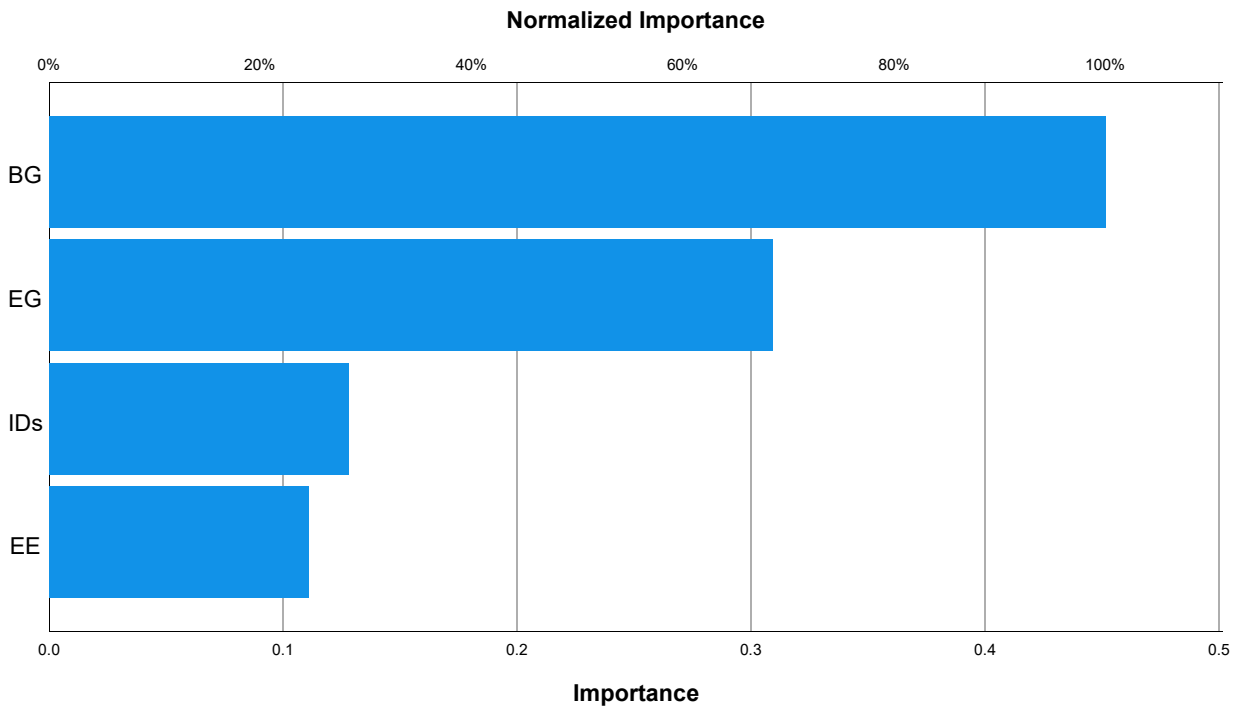
Training	Sum of Squares Error	206.054
	Relative Error	.785
	Stopping Rule Used	1 consecutive step (s) with no decrease in error ^a
	Training Time	0:00:00.08
Testing	Sum of Squares Error	78.609
	Relative Error	.713

Dependent Variable: CD

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
IDs	.128	28.4%
EE	.111	24.5%
BG	.452	100.0%
EG	.309	68.5%



Multilayer Perceptron

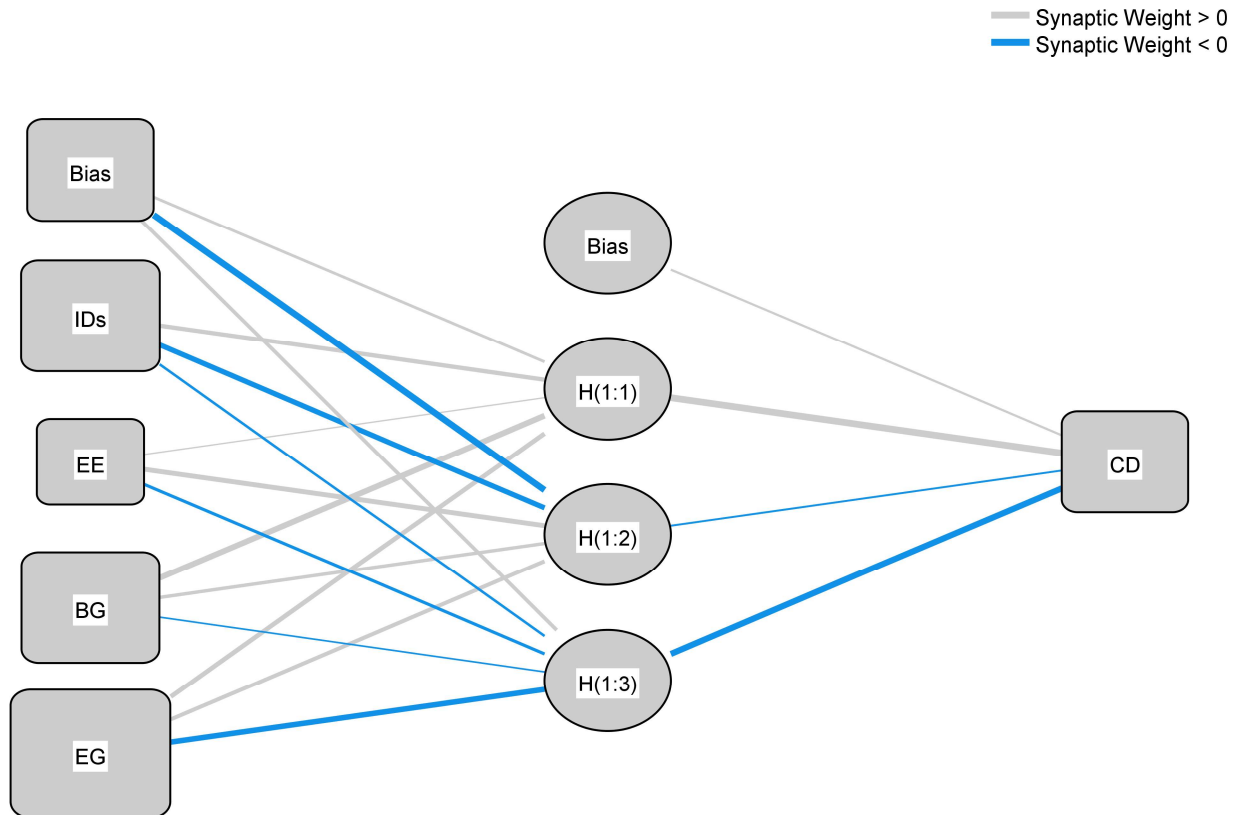
Case Processing Summary

		N	Percent
Sample	Training	503	67.1%
	Testing	247	32.9%
Valid		750	100.0%
Excluded		0	
Total		750	

Network Information

Input Layer	Covariates	1	IDs
		2	EE
		3	BG
		4	EG
	Number of Units ^a		4
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1 ^a		3
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	CD
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit



Hidden layer activation function: Hyperbolic tangent

Output layer activation function: Identity

Model Summary

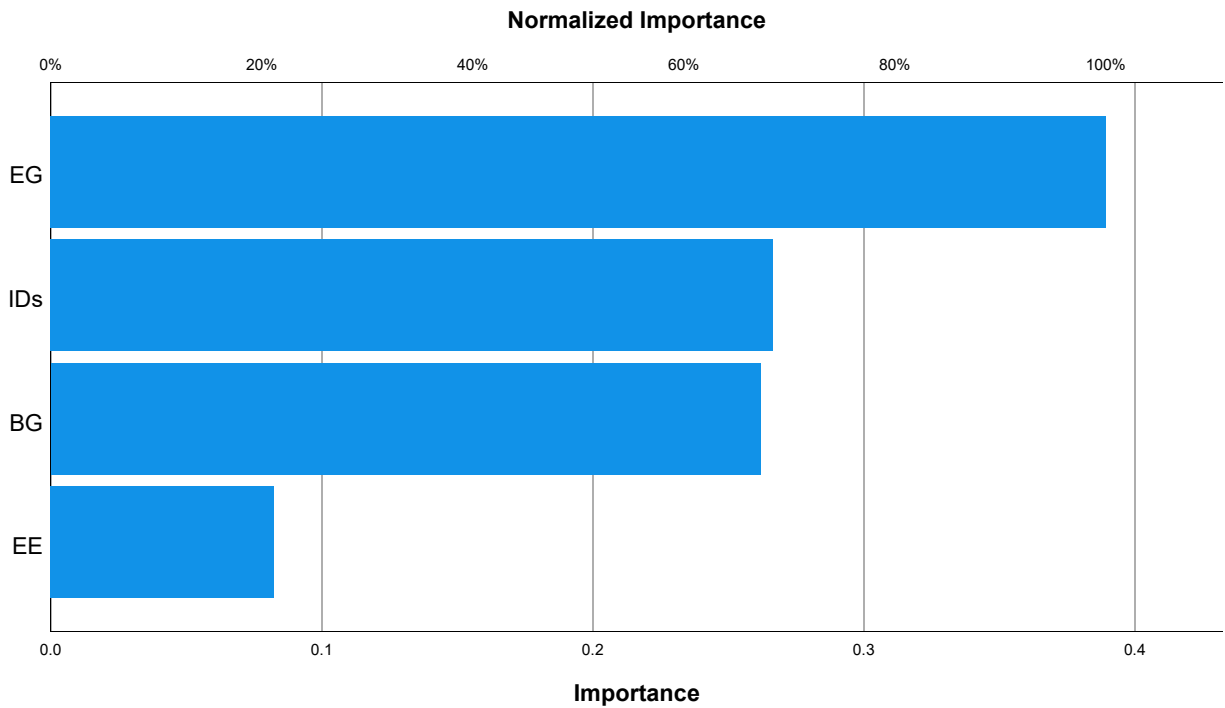
Training	Sum of Squares Error	186.953
	Relative Error	.745
	Stopping Rule Used	1 consecutive step (s) with no decrease in error ^a
	Training Time	0:00:00.06
Testing	Sum of Squares Error	86.494
	Relative Error	.811

Dependent Variable: CD

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
IDs	.266	68.4%
EE	.082	21.2%
BG	.262	67.2%
EG	.389	100.0%



Multilayer Perceptron

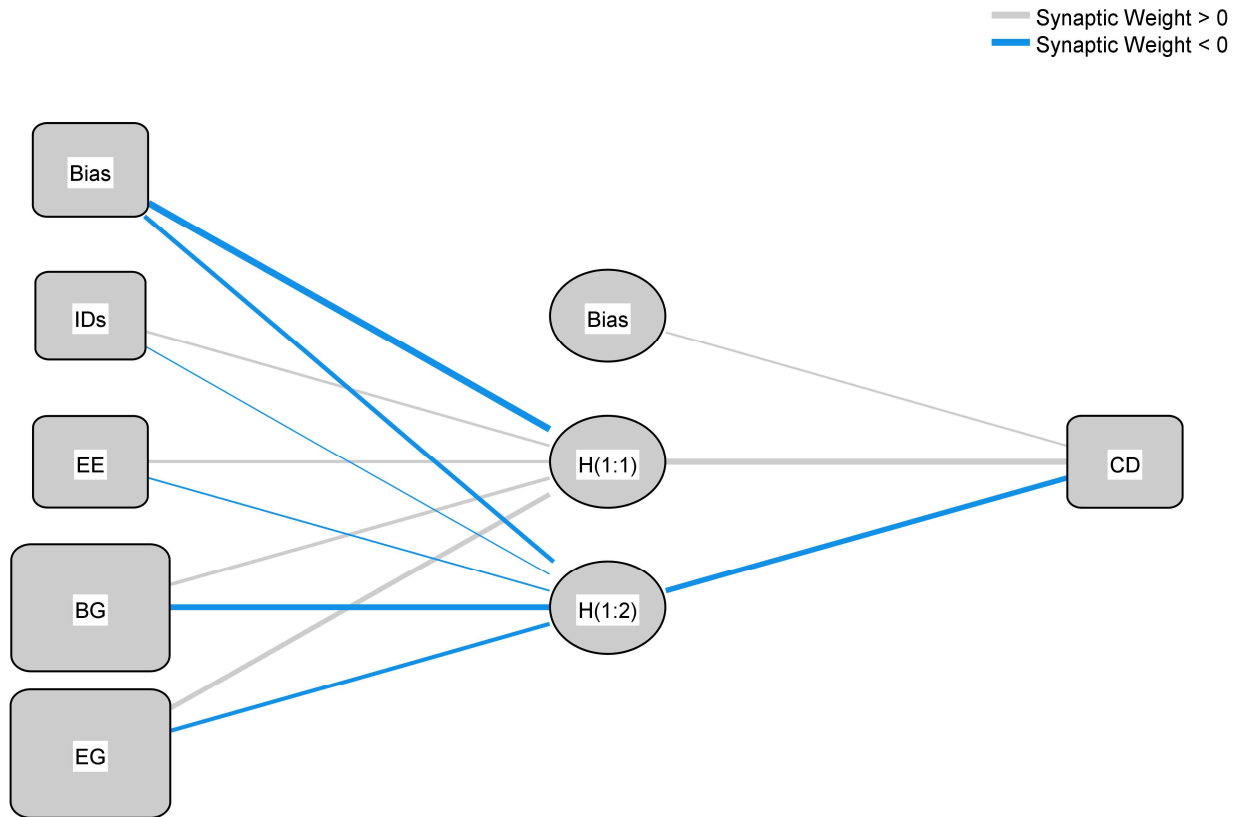
Case Processing Summary

		N	Percent
Sample	Training	522	69.6%
	Testing	228	30.4%
Valid		750	100.0%
Excluded		0	
Total		750	

Network Information

Input Layer	Covariates	1	IDs
		2	EE
		3	BG
		4	EG
	Number of Units ^a		4
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1 ^a		2
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	CD
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit



Hidden layer activation function: Hyperbolic tangent

Output layer activation function: Identity

Model Summary

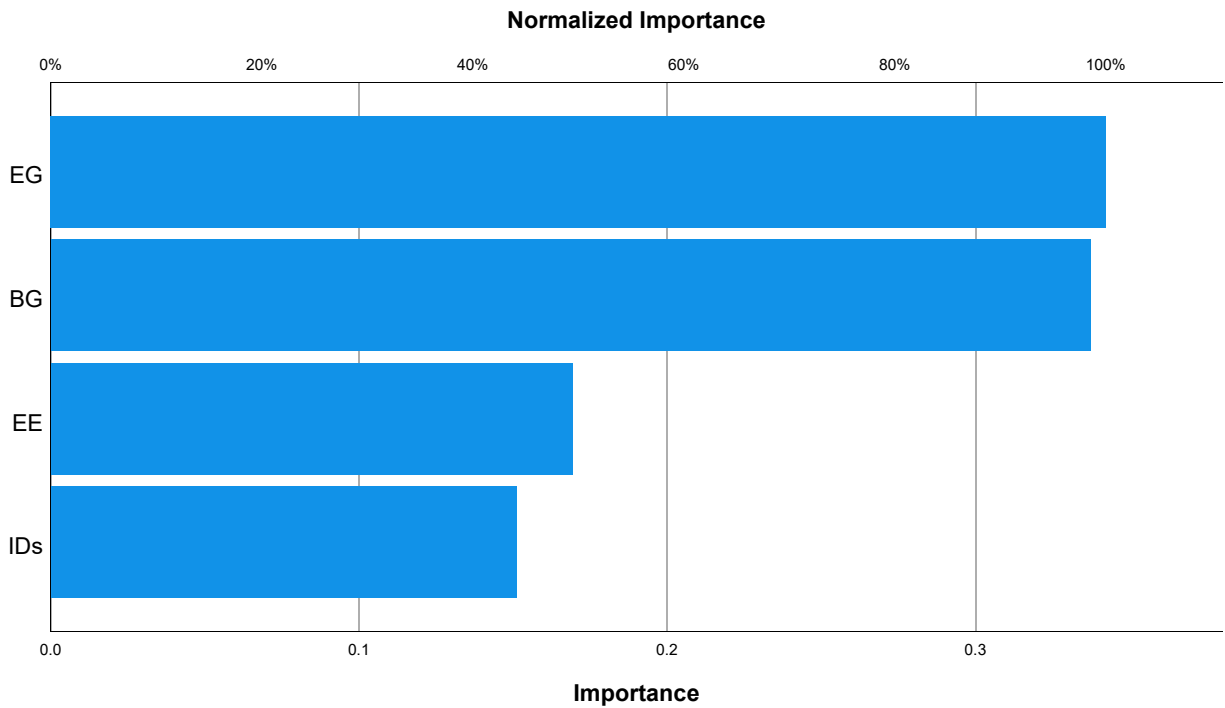
Training	Sum of Squares Error	199.863
	Relative Error	.767
	Stopping Rule Used	1 consecutive step (s) with no decrease in error ^a
	Training Time	0:00:00.06
Testing	Sum of Squares Error	90.968
	Relative Error	.715

Dependent Variable: CD

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
IDs	.151	44.1%
EE	.169	49.4%
BG	.337	98.5%
EG	.342	100.0%



Multilayer Perceptron

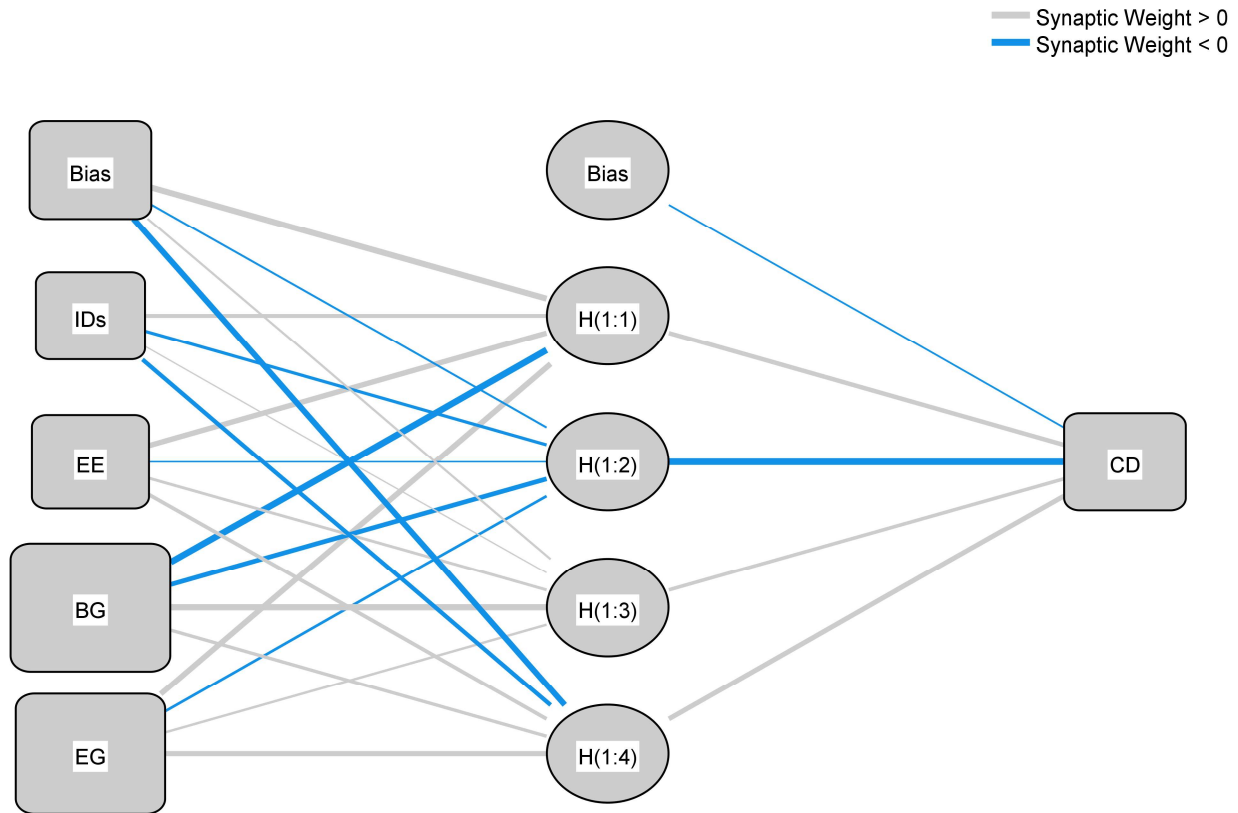
Case Processing Summary

		N	Percent
Sample	Training	518	69.1%
	Testing	232	30.9%
Valid		750	100.0%
Excluded		0	
Total		750	

Network Information

Input Layer	Covariates	1	IDs
		2	EE
		3	BG
		4	EG
	Number of Units ^a		4
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1 ^a		4
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	CD
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit



Hidden layer activation function: Hyperbolic tangent

Output layer activation function: Identity

Model Summary

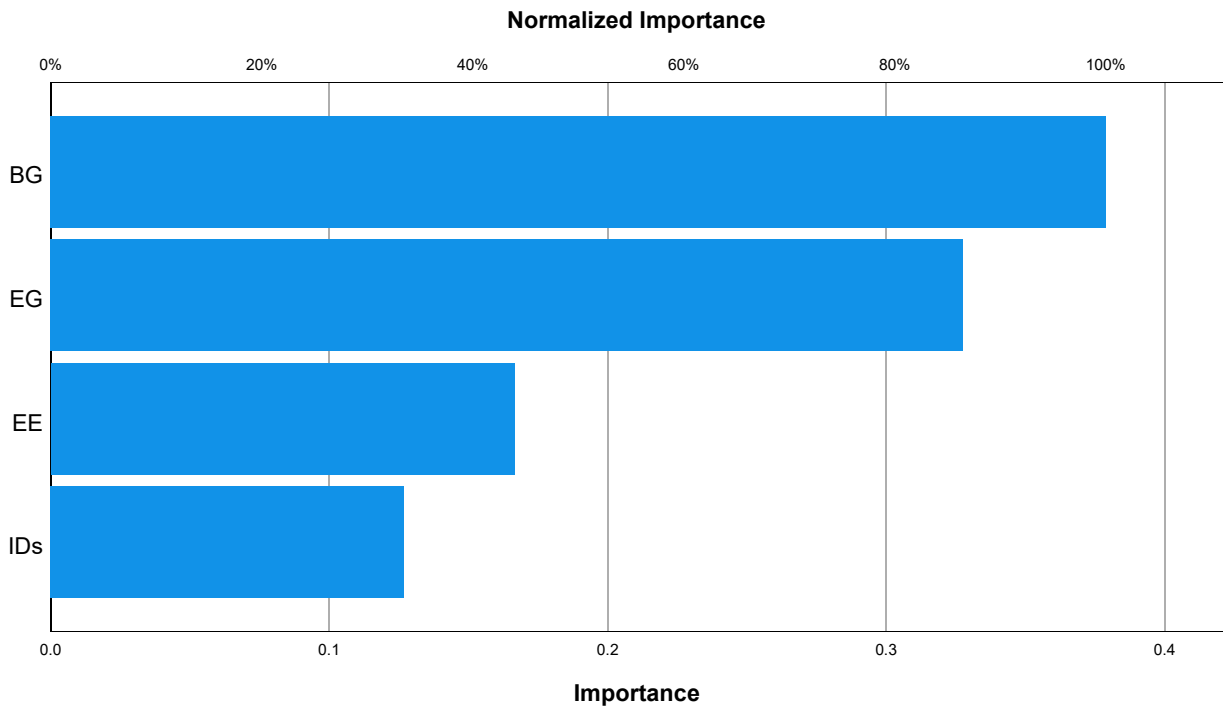
Training	Sum of Squares Error	194.448
	Relative Error	.752
	Stopping Rule Used	1 consecutive step (s) with no decrease in error ^a
	Training Time	0:00:00.13
Testing	Sum of Squares Error	82.879
	Relative Error	.732

Dependent Variable: CD

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
IDs	.127	33.5%
EE	.167	44.0%
BG	.379	100.0%
EG	.328	86.4%



Multilayer Perceptron

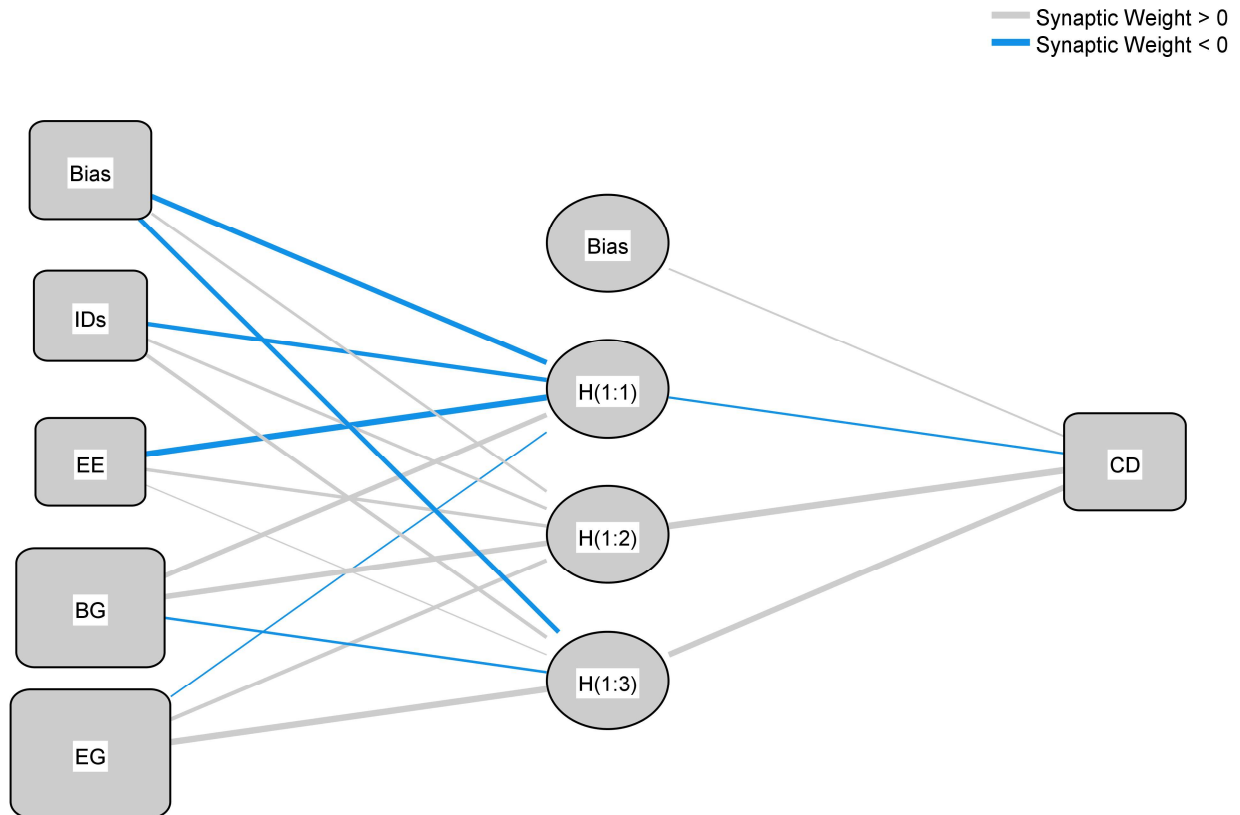
Case Processing Summary

		N	Percent
Sample	Training	535	71.3%
	Testing	215	28.7%
Valid		750	100.0%
Excluded		0	
Total		750	

Network Information

Input Layer	Covariates	1	IDs
		2	EE
		3	BG
		4	EG
	Number of Units ^a		4
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1 ^a		3
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	CD
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit



Hidden layer activation function: Hyperbolic tangent

Output layer activation function: Identity

Model Summary

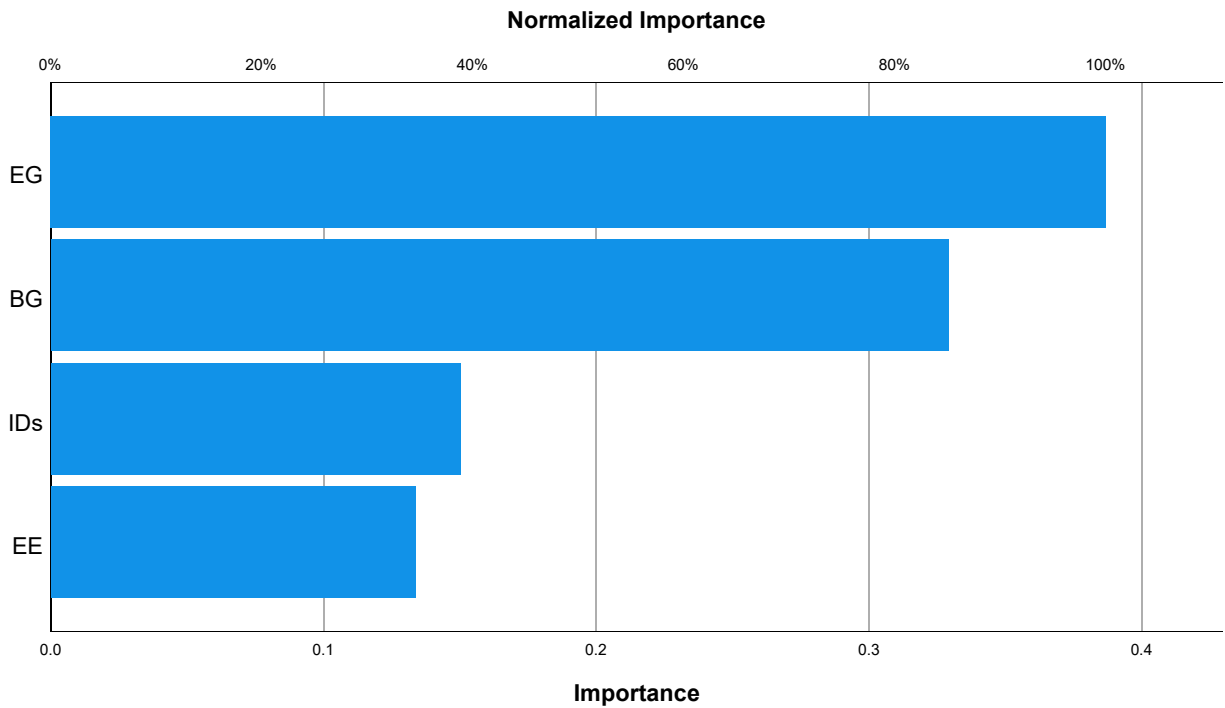
Training	Sum of Squares Error	196.948
	Relative Error	.738
	Stopping Rule Used	1 consecutive step (s) with no decrease in error ^a
	Training Time	0:00:00.05
Testing	Sum of Squares Error	83.878
	Relative Error	.778

Dependent Variable: CD

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
IDs	.150	38.8%
EE	.134	34.6%
BG	.329	85.1%
EG	.387	100.0%



Multilayer Perceptron

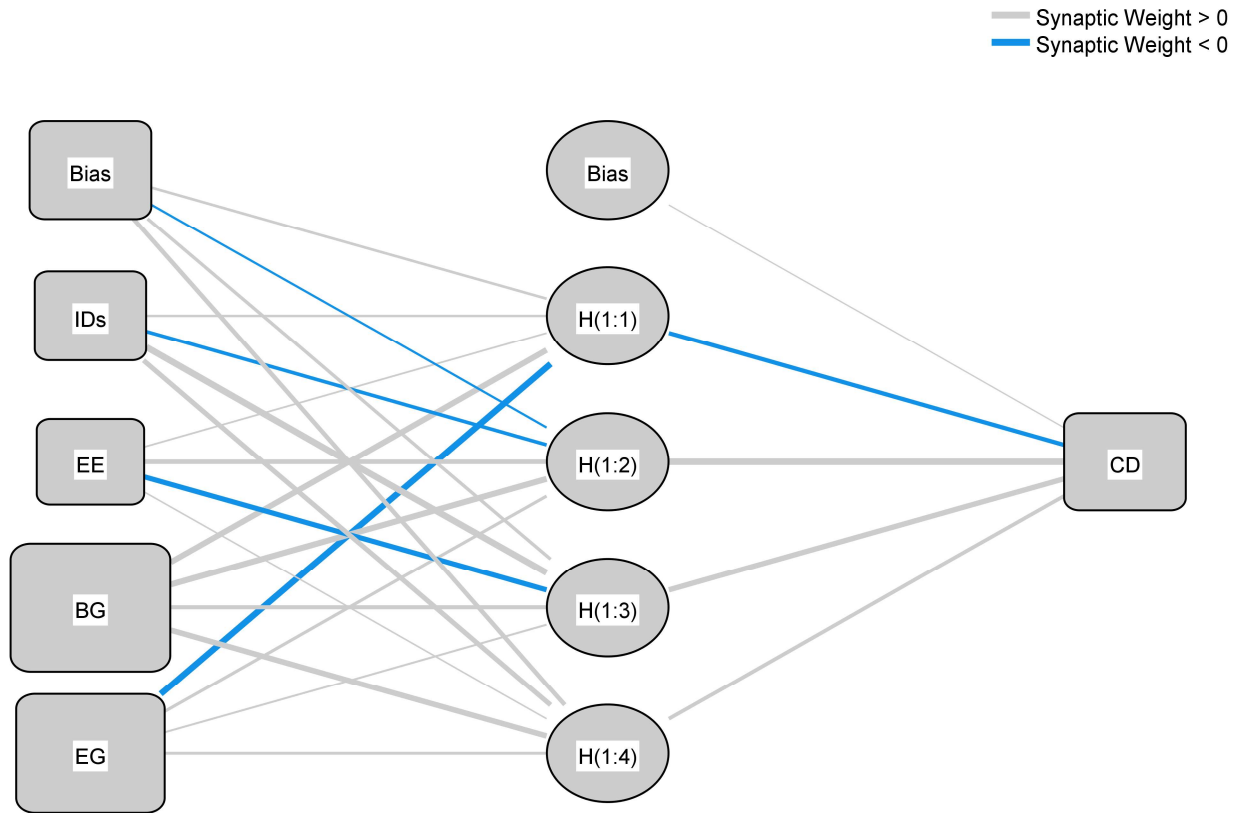
Case Processing Summary

		N	Percent
Sample	Training	525	70.0%
	Testing	225	30.0%
Valid		750	100.0%
Excluded		0	
Total		750	

Network Information

Input Layer	Covariates	1	IDs
		2	EE
		3	BG
		4	EG
	Number of Units ^a		4
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1 ^a		4
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	CD
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit



Hidden layer activation function: Hyperbolic tangent

Output layer activation function: Identity

Model Summary

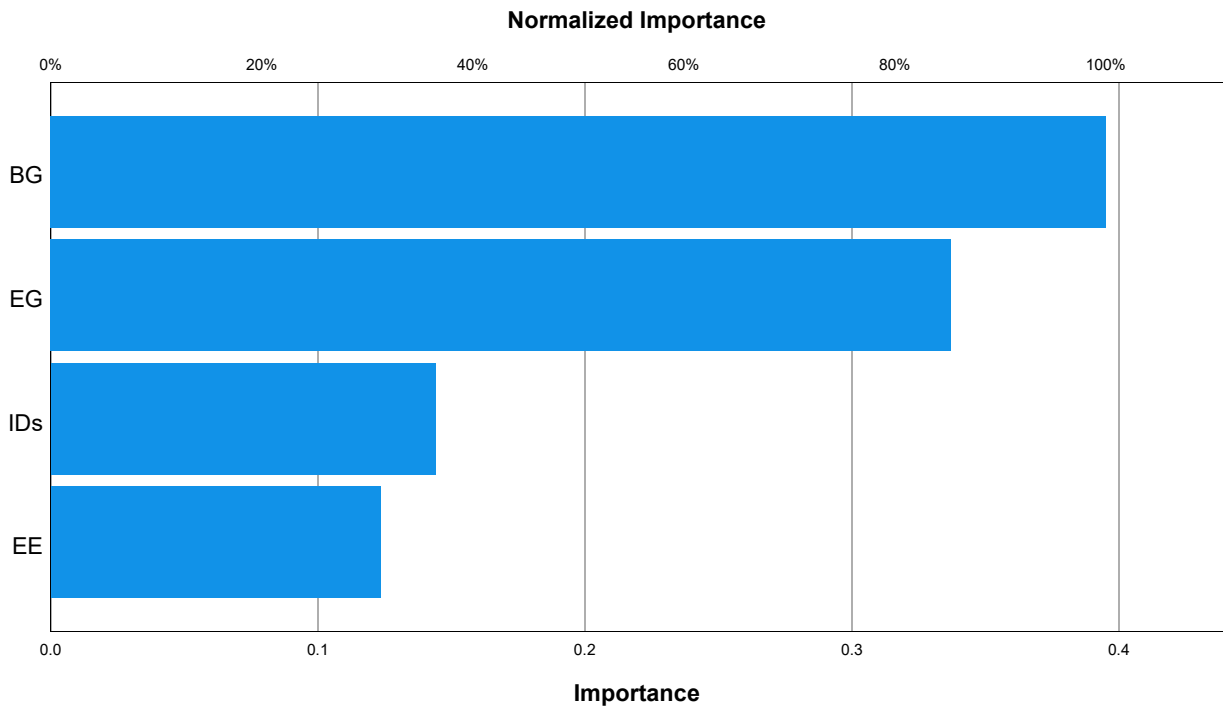
Training	Sum of Squares Error	195.564
	Relative Error	.746
	Stopping Rule Used	1 consecutive step (s) with no decrease in error ^a
	Training Time	0:00:00.06
Testing	Sum of Squares Error	82.278
	Relative Error	.737

Dependent Variable: CD

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
IDs	.144	36.5%
EE	.123	31.2%
BG	.395	100.0%
EG	.337	85.3%



Multilayer Perceptron

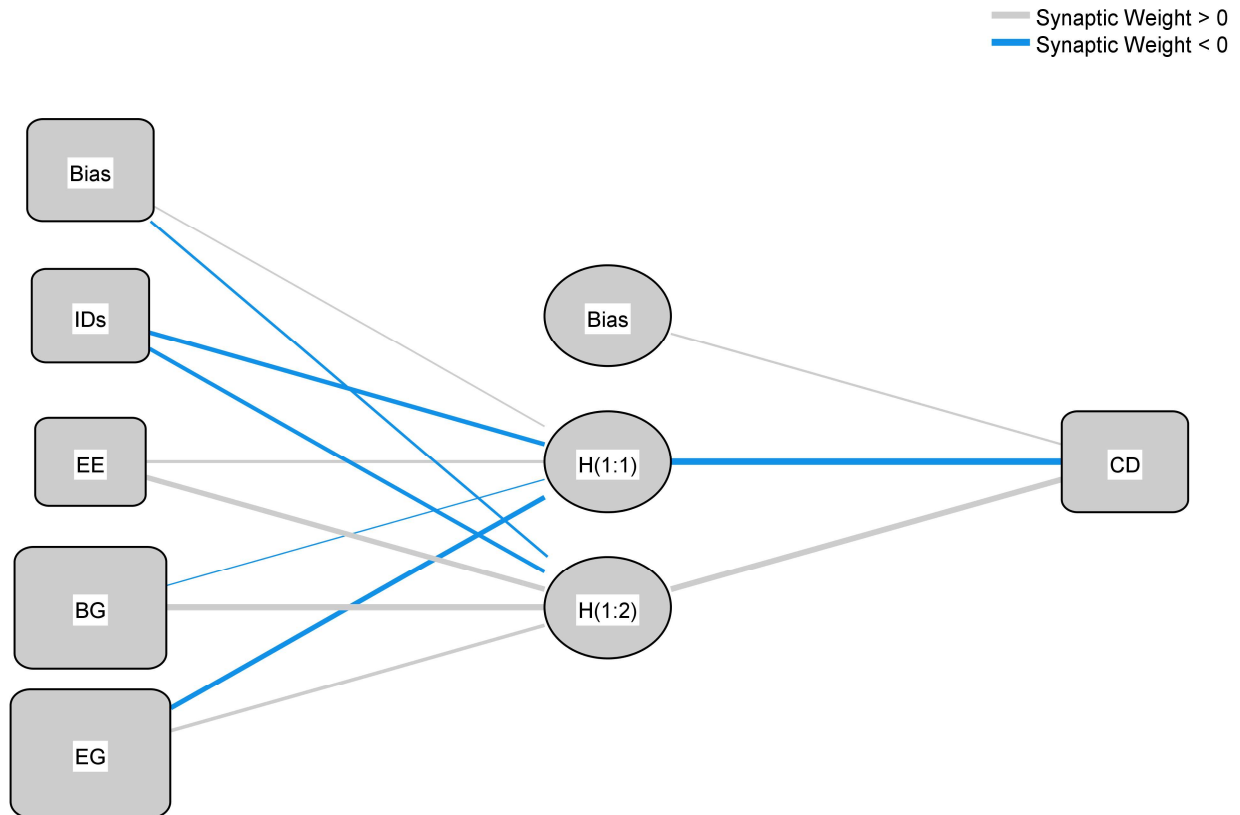
Case Processing Summary

		N	Percent
Sample	Training	526	70.1%
	Testing	224	29.9%
Valid		750	100.0%
Excluded		0	
Total		750	

Network Information

Input Layer	Covariates	1	IDs
		2	EE
		3	BG
		4	EG
	Number of Units ^a		4
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1 ^a		2
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	CD
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit



Hidden layer activation function: Hyperbolic tangent

Output layer activation function: Identity

Model Summary

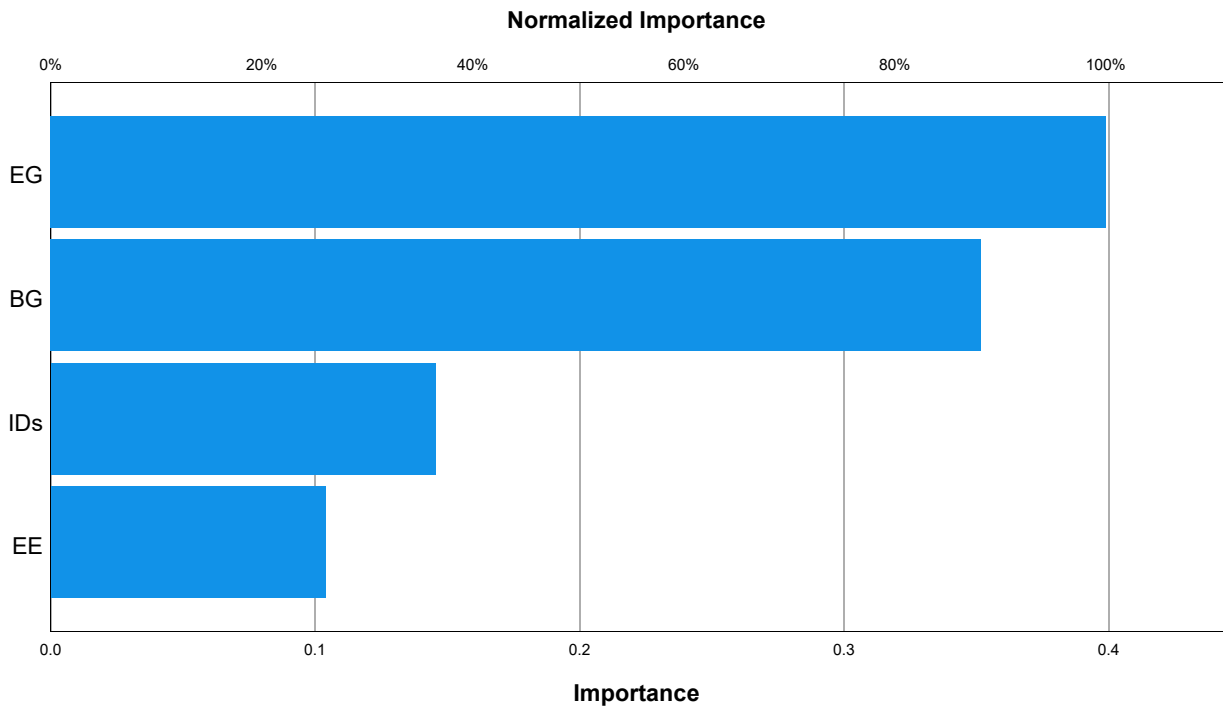
Training	Sum of Squares Error	202.202
	Relative Error	.770
	Stopping Rule Used	1 consecutive step (s) with no decrease in error ^a
	Training Time	0:00:00.06
Testing	Sum of Squares Error	83.741
	Relative Error	.698

Dependent Variable: CD

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
IDs	.145	36.4%
EE	.104	26.0%
BG	.352	88.1%
EG	.399	100.0%



Multilayer Perceptron

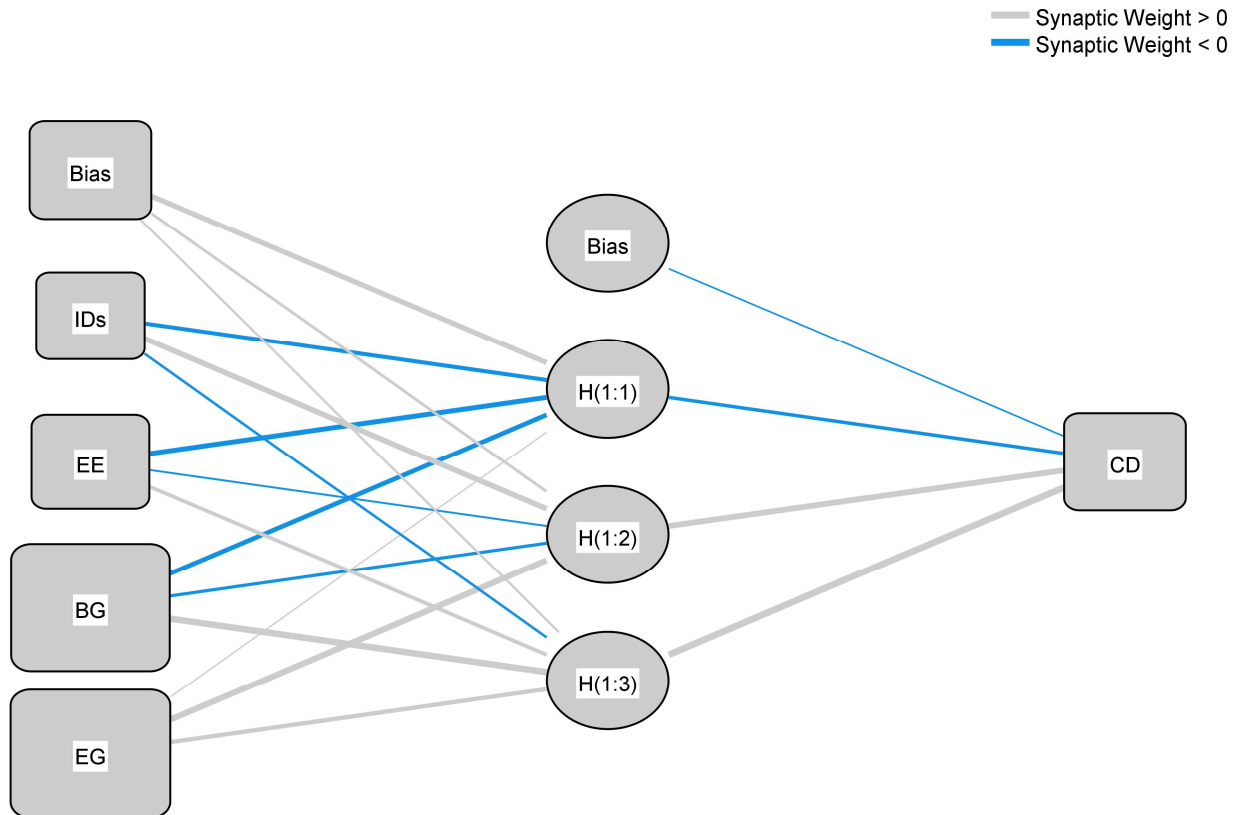
Case Processing Summary

		N	Percent
Sample	Training	532	70.9%
	Testing	218	29.1%
Valid		750	100.0%
Excluded		0	
Total		750	

Network Information

Input Layer	Covariates	1	IDs
		2	EE
		3	BG
		4	EG
	Number of Units ^a		4
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1 ^a		3
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	CD
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit



Hidden layer activation function: Hyperbolic tangent

Output layer activation function: Identity

Model Summary

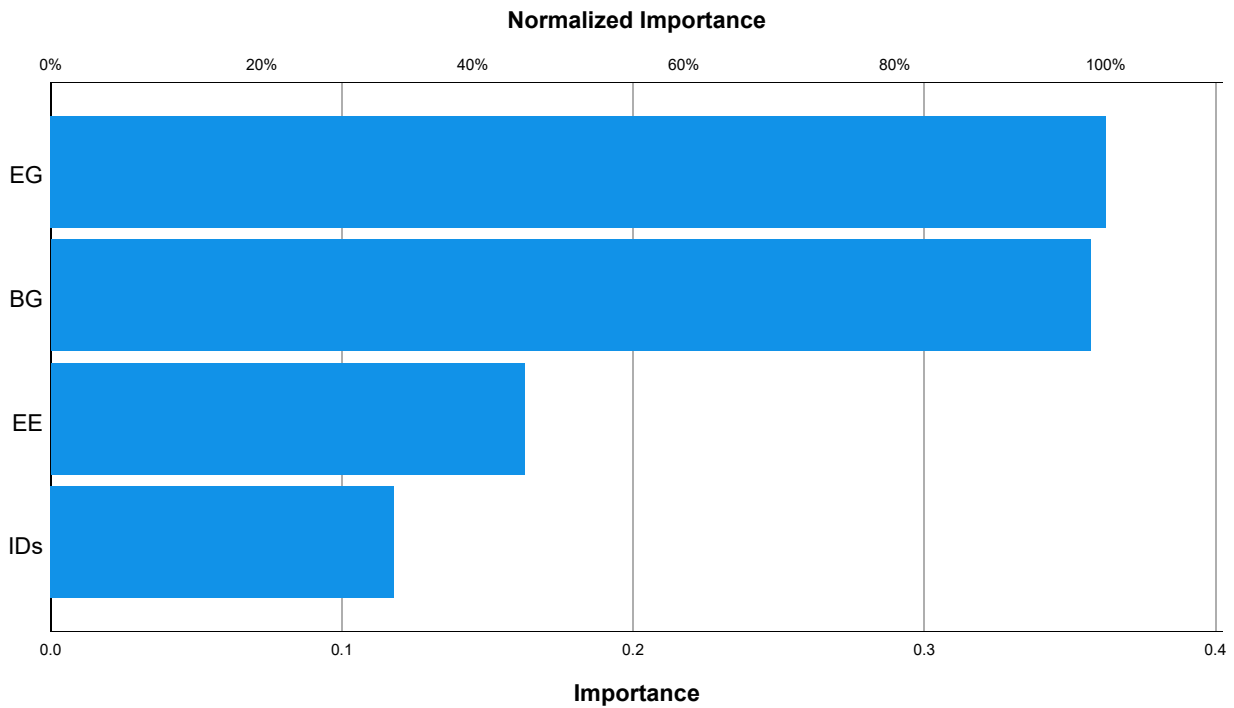
Training	Sum of Squares Error	199.535
	Relative Error	.752
	Stopping Rule Used	1 consecutive step (s) with no decrease in error ^a
	Training Time	0:00:00.05
Testing	Sum of Squares Error	78.104
	Relative Error	.746

Dependent Variable: CD

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
IDs	.118	32.5%
EE	.163	44.9%
BG	.357	98.5%
EG	.362	100.0%



Multilayer Perceptron

Case Processing Summary

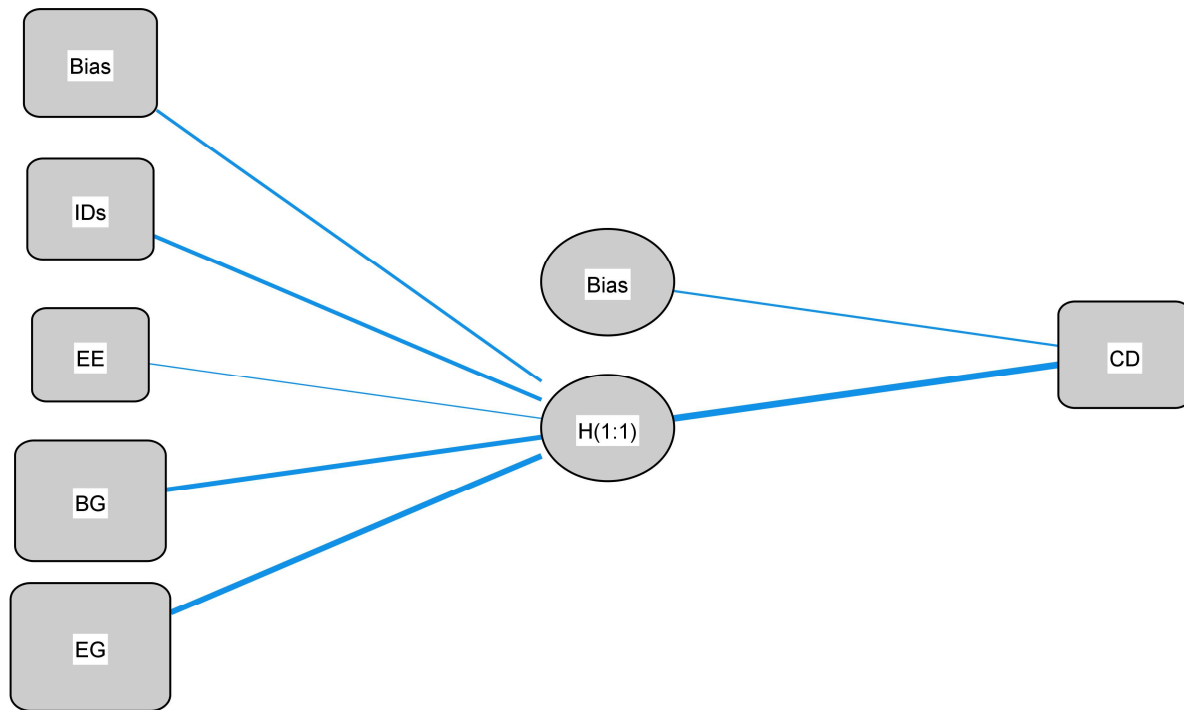
		N	Percent
Sample	Training	533	71.1%
	Testing	217	28.9%
Valid		750	100.0%
Excluded		0	
Total		750	

Network Information

Input Layer	Covariates	1	IDs
		2	EE
		3	BG
		4	EG
	Number of Units ^a		4
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1 ^a		1
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	CD
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit

— Synaptic Weight < 0



Hidden layer activation function: Hyperbolic tangent

Output layer activation function: Identity

Model Summary

Training	Sum of Squares Error	206.543
	Relative Error	.776
	Stopping Rule Used	1 consecutive step (s) with no decrease in error ^a
	Training Time	0:00:00.05
Testing	Sum of Squares Error	75.021
	Relative Error	.752

Dependent Variable: CD

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
IDs	.155	37.2%
EE	.080	19.2%
BG	.347	83.1%
EG	.417	100.0%

